Exact Recovery of a Linear Differential Operator from a Single Solution

Volume I: The Theory, Phases 0–8

October 2026

Abstract

Given one exact solution $u(t, x)$ of a differential equation in one space and one time variable, what does u tell us about the equation? This volume collects the whole theory built to answer that question, from the first constant-coefficient results (identification, factor decomposition, the interaction term, a nonlinear extension) to the later phases: systems and gauge, variable coefficients and potentials, an analytic encoder, a universal dictionary with misfit certificates, convergence under noise, locality, dimension and boundary conditions. Each result carries a trust label, each chapter begins with its assumptions, purpose and role, and the last chapters state what is false, what is open and what has not been done. Proofs are in Volume II.

Contents

1 Orientation **4**

1.1 The three questions and a fourth . . . 4

1.2 How to read a statement: the trust labels . . . 4

1.3 Map of the dependencies . . . 5

1.4 What each chapter contains . . . 6

1.5 Notation used throughout . . . 6

2 The core in brief **7**

2.1 Derivatives are polynomials . . . 7

2.2 What a solution "remembers" . . . 7

2.3 The four results of the short paper . . . 8

3 Identification **9**

3.1 The criterion . . . 9

3.2 The intrinsic operator and the spectral form . . . 10

3.3 How much data is enough? . . . 10

3.4 What no finite data can show . . . 10

3.5 Variable coefficients: first limits . . . 11

3.6 What is not proved . . . 11

4 Decomposition **12**

4.1 Existence, ambiguity, and the topology of Ω . . . 12

4.2 Explicit projectors . . . 13

4.3 Regularity: when the pieces stay in H^m . . . 13

4.4 Variable coefficients . . . 14

5 Terms inside a factor, and the interaction term **15**

5.1 Definition and exact formula . . . 15

5.2 Cost, and an instability . . . 16

5.3 Time-dependent perturbation, boundaries, non-commuting terms . . . 16

6 Nonlinear equations **18**

6.1 The nonlinear interaction term (A9) . . . 18

6.2 Across the shock . . . 18

6.3 Identification relative to a dictionary . . . 19

6.4 Integrable models, scattering, analyticity . . . 20

7 Systems of equations and gauge **21**

7.1 When does one solution identify a system? . . . 21

7.2 Sum of kernels for systems . . . 22

7.3 Interaction for systems, and the Dirac case . . . 22

7.4 Gauge . . . 23

7.5 Open and false . . . 23

8 Variable coefficients: identifying a potential **24**

8.1 Identifying V itself . . . 24

8.2 ODEs: structure and Galois theory . . . 25

8.3 Dynamical Lie algebras, Pöschl–Teller, non-autonomous . . . 26

9 The analytic encoder: removing a potential **27**

9.1 What is an encoder? . . . 27

9.2 The Volterra transformation operator . . . 27

9.3 What cannot exist . . . 29

9.4 The interaction term for a potential . . . 29

10 A universal dictionary and misfit certificates **30**

10.1 Universality: one class at a time . . . 31

10.2 The negative theorem N1: what "everything" cannot mean . . . 31

10.3 Misfit certificates . . . 32

10.4 Best approximation and the rules R0–R6 . . . 32

11 Convergence, noise, locality, dimension, boundary **34**

11.1 Identification in the limit . . . 34

11.2 Noisy data . . . 35

11.3 Locality . . . 35

11.4 Dimension $1 + d'$ . . . 35

11.5 Boundary conditions . . . 36

11.6 Splitting schemes with a boundary . . . 36

11.7 Mori–Zwanzig . . . 36

11.8 The convergence contract for the algorithm . . . 36

12 The expanded algorithm **38**

12.1 The flow . . . 38

12.2 Step by step . . . 38

12.3 What the output can honestly claim 39

13 Status, gaps, and what remains **41**

13.1 Scoreboard 41
13.2 What “independently checked” meant 42
13.3 The formalised core (Lean 4 / Mathlib) 42
13.4 Textbook sanity checks 42
13.5 Statements the programme proved false 43
13.6 Open and conditional 43
13.7 What these volumes do not contain 44

Sources cited **45**

CHAPTER 1

Orientation

In this chapter	
Starting from	One exact solution $u(t,x)$ of a linear or nonlinear PDE in one space and one time variable, on an open set Ω. The reader has seen the short paper ("Exact Recovery of a Linear Differential Operator from a Single Solution"); this volume is self-contained but compresses what that paper explains at length.
Why this chapter	The research programme grew through eleven phases. The short paper stops at Phase 5 and states results without proofs. This volume gives the whole programme up to Phase 8 in one connected narrative, and Volume II gives the proofs.
Role in the volume	A map of the whole: what the questions are, what each chapter answers, how much trust each answer deserves, and what is *not* done.

1.1 The three questions and a fourth

Given one function u known to solve some equation $L(u) = 0$:

Identify Can u alone tell us the equation? (Chapters 3, 7, 8)

Decompose Can u be split into pieces, one per part of the equation? (Chapter 4)

Interact If the equation is a sum of terms, how much of u is due to the terms *interacting*? (Chapters 5, 6)

Extend What changes for systems, variable coefficients, noise, nonlocal laws, and when the true law is not in our catalogue? (Chapters 7–11)



1.2 How to read a statement: the trust labels

Every result carries one label. They are used identically in both volumes.

Label	Meaning
[proved]	A complete proof exists and an independent reviewer (a fresh AI agent, not a human referee) checked it, with an adversarial pass on the main results.
[proof written, not audited]	A complete proof is written in the research files but no independent check has been done. Treat as likely correct, not as settled.
[from cited source]	A published theorem whose source was checked; we use it, we do not reprove it.
[proved mod. X]	Proved assuming a named published fact or an assumption X.
[conditional]	Depends on an unverified or stated hypothesis.
[checked on examples only]	Verified on examples by computer algebra only.
[open]	Not known.
[false]	A counterexample exists. Included because they shape the theory.

What "proved" does and does not mean here

The reviewers were independent AI agents (fresh context, no access to the author's reasoning), not human referees. Fifteen purely algebraic statements were also formalised in Lean 4/Mathlib. Nothing has been peer-reviewed. Where a ledger row says "proposed state" (Phase 8), the registration in the research ledger is still pending.

1.3 Map of the dependencies

Figure 1.1 shows how the load-bearing theorems depend on each other. It is not drawn by hand: arrows are the `depends_on` relations extracted from the project's knowledge graph (2197 nodes, built from the ledger, proof packages and audits). Foundations are at the top.

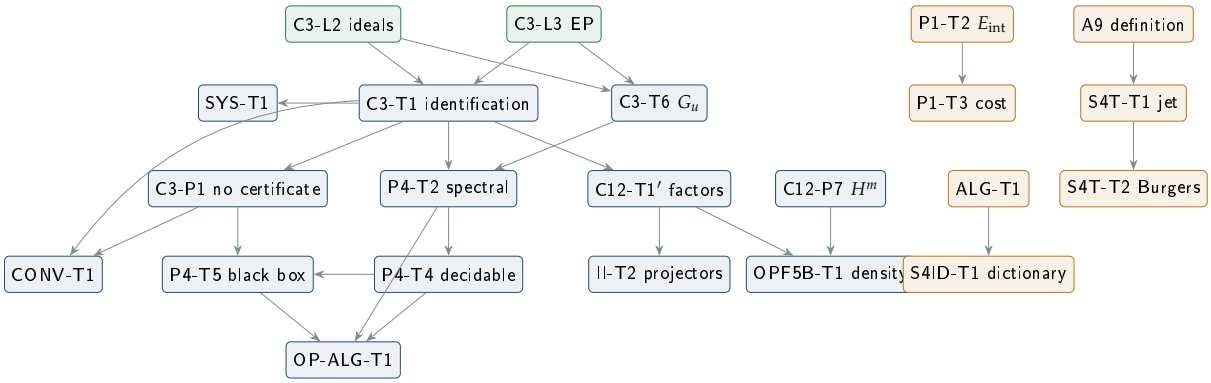


Figure 1.1: Dependencies among load-bearing results. Green: algebraic foundations; blue: linear theory; orange: interaction and nonlinear theory. An arrow $X \to Y$ means Y uses X. The orange cluster is almost independent of the blue one: *interaction* needs only commuting flows, not the identification machinery.

1.4 What each chapter contains

Ch.	Topic	Main answer
2	Core in brief	One-page recap of the ring, annihilators, factors, interaction.
3	Identification	Exactly when u determines P; finite data, undecidability, special families.
4	Decomposition	Splitting per factor on simply connected sets; holes; regularity.
5	Terms & E_{int}	Closed form, cost, instability, Lie classification, boundary, regularisation limits.
6	Nonlinear	Jets, Burgers shocks, dictionaries, KdV and scattering.
7	Systems & gauge	Matrix coefficients: when a solution identifies, up to gauge.
8	Variable coefficients	Identify a potential $V(x)$ or $V(t,x)$; Galois structure.
9	Analytic encoder	Remove V by transformation operators; what does not exist.
10	Universal dictionary	Which operators a countable catalogue can approximate; misfit certificates.
11	Convergence	Limits, noise, locality, dimension, boundary, splitting.
12	Expanded algorithm	Seven steps, each with its guarantee.
13	Status & gaps	What is proved, what is open, what is left of the plan.

1.5 Notation used throughout

Symbol	Meaning
$\mathcal{R} = \mathbb{C}[\tau, \xi]$	Polynomial ring; $\tau \leftrightarrow \partial_t$, $\xi \leftrightarrow \partial_x$. An operator is an element of $\mathcal{R}$.
$V(I) \subset \mathbb{C}^2$	Common zeros of the ideal I; for P the *characteristic variety*.
$\mathrm{Ann}(u)$	Ideal of operators annihilating u.
$G_u,\ J_u$	$\mathrm{Ann}(u) = G_u J_u$, $G_u = \gcd$, $V(J_u)$ finite (Chapter 3).
$\mathrm{EP}(\Omega)$	Exponential-polynomials $\sum q_k(t,x) e^{\langle z_k, (t,x)\rangle}$.
$\delta(P)$	Dimension of the ambiguity of the factor decomposition.
E_{int}	Interaction term.
$\mathcal{D}_n, \mathcal{D}_\infty$	Level-n and full dictionary of candidate operators (Chapter 10).
$\sim$	Equal up to a nonzero constant.

CHAPTER 2

The core in brief

In this chapter	
Starting from	Constant-coefficient linear PDEs in (t, x), one exact solution known on an open connected set $\Omega \subset \mathbb{R}^2$.
Why this chapter	The later chapters extend, sharpen or break the picture drawn here. Readers of the short paper can skip to Chapter 3; others should read this once.
Role in the volume	Fixes the vocabulary: operator = polynomial, solution = element of a module, component = factor, term = summand.

2.1 Derivatives are polynomials

Write τ for ∂_t and ξ for ∂_x. A constant-coefficient operator such as $\partial_t - D\partial_x^2$ is the polynomial $P = \tau - D\xi^2$ in $\mathcal{R} = \mathbb{C}[\tau, \xi]$. The exponential $e^{\sigma t + \zeta x}$ is an eigenfunction of every derivative: $\partial_t e^{\sigma t+\zeta x} = \sigma e^{\sigma t+\zeta x}$. Hence

$$P(\partial)\, e^{\sigma t+\zeta x} = P(\sigma, \zeta)\, e^{\sigma t+\zeta x},$$

and $e^{\sigma t+\zeta x}$ solves $P(\partial)u = 0$ exactly when (σ, ζ) lies on the *characteristic variety* $V(P) = \{P = 0\} \subset \mathbb{C}^2$.

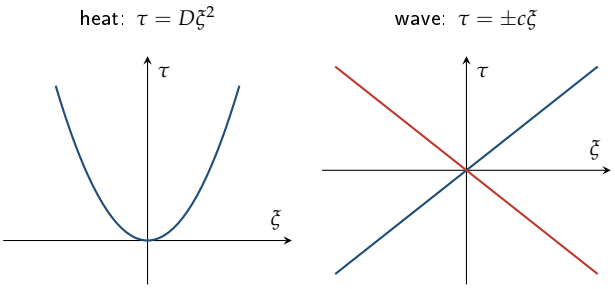


Figure 2.1: Real slices of the characteristic variety. Heat: one irreducible curve. Wave: two lines (two factors) crossing at the origin, which is why the decomposition into $f(x - ct)$ and $g(x + ct)$ is ambiguous by a constant.

2.2 What a solution "remembers"

Definition 2.1 Annihilator and intrinsic operator

$\mathrm{Ann}(u) = \{Q \in \mathcal{R} : Q(\partial)u = 0\}$, an ideal. Write $G_u = \gcd \mathrm{Ann}(u)$ and $\mathrm{Ann}(u) = G_u \cdot J_u$ with $V(J_u)$ finite (Volume II, Chapter 2, ideal structure theorem).

The ideal $\mathrm{Ann}(u)$ is everything u knows about constant-coefficient laws. If $P(\partial)u = 0$ we say u *identifies* P when $\mathrm{Ann}(u) = (P)$: then every other law satisfied by u is a consequence of P.

2.3 The four results of the short paper

1. **Identification** [proved]. $\mathrm{Ann}(u) = (P)$ iff no partial quotient $(P/P_i)(\partial)u$ is an exponential-polynomial; equivalently $\sigma_p(u) = \emptyset$ and $G_u \sim P$. No finite amount of data certifies it, a black-box test is impossible, a single bound B makes it decidable (Chapter 3).
2. **Decomposition** [proved]. On simply connected Ω the solution splits into one piece per irreducible factor; the ambiguity is an exponential-polynomial space of dimension $\delta(P)$; it fails with holes; in H^m it holds iff the sum of kernels is closed and dense (Chapter 4).
3. **Interaction** [proved]. For commuting terms $E_{\mathrm{int}} = \prod_n(\mathrm{Id} + \Delta_n) - \mathrm{Id} - \sum_n \Delta_n$ applied to u_0, cost $O(N)$ per value, $\Theta(2^N)$ as a full table, not continuous in H^m (Chapter 5).
4. **Nonlinear** [proved]/[proved mod. Kružkov]. Jet expansions, Burgers before and after the shock, dictionary identification (Chapter 6).

> **Takeaway.** The three words "operator, component, term" mean three different algebraic objects: an operator is a polynomial, a component is a *factor* of it (not a summand), a term is a *summand*. Splitting per factor works; splitting per term fails in general, and what is left over is E_{int}.

CHAPTER 3

Identification

In this chapter

Starting from	$u \in \mathcal{D}'(\Omega)$ on a connected open $\Omega \subset \mathbb{R}^2$ with $P(\partial)u = 0$, $P = c\prod P_i^{e_i}$ nonconstant, constant coefficients. Proofs: Volume II, Chapters 2 and 3.
Why this chapter	Before decomposing a solution one must know the equation. Here we ask exactly when the single function u pins down P, how much data is enough, and what no finite data can ever show.
Role in the volume	First link of the chain identify → decompose → interact. Its output is the pair (G_u, J_u) that all later chapters use.



3.1 The criterion

Theorem 3.1 Identification criterion

Let $w_i = (P/P_i)(\partial)u$. Then $\mathrm{Ann}(u) = (P)$ *iff* no w_i is an exponential-polynomial. [proved] (C3-T1)

In words: u fails to identify P precisely when, after removing one irreducible factor, what is left of u is a finite sum of $q(t,x)e^{\sigma t+\zeta x}$. Such a leftover is a "finite-dimensional" object; it is killed by operators that P knows nothing about.

Intuition

The curve $V(P)$ has infinitely many points. A function u "uses" infinitely many of them unless some factor is concentrated on finitely many points, i.e. unless the part of u belonging to that factor is an exponential-polynomial. Figure 3.1 shows both situations on the heat curve.

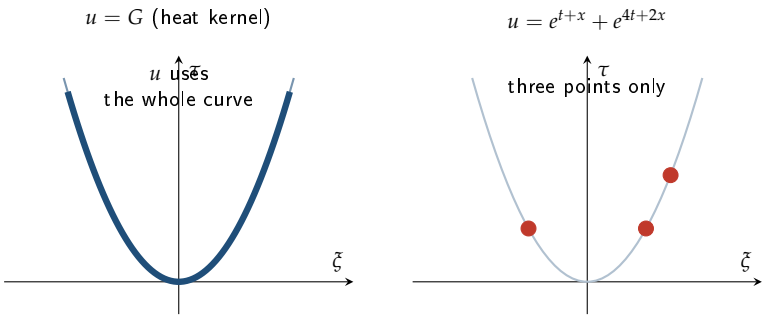


Figure 3.1: Left: a solution spread over the characteristic curve; $\mathrm{Ann}(u) = (P)$. Right: a finite sum of exponentials lives on finitely many points $z_k \in V(P)$, so $\mathrm{Ann}(u) \supsetneq (P)$ (it contains polynomials vanishing at the z_k that P does not divide).

3.2 The intrinsic operator and the spectral form

$\mathrm{Ann}(u) = G_u \cdot J_u$; G_u is the *operator that u really satisfies*, J_u the finite leftover.

Theorem 3.2 Intrinsic operator

If $P \in \mathrm{Ann}(u)$ is any annihilator, $P = c\prod P_i^{e_i}$, then $G_u \sim \prod P_i^{f_i}$ with $f_i = e_i - \max\{k : (P/P_i^k)(\partial)u \in \mathrm{EP}\}$. $\mathrm{Ann}(u)$ is principal iff $G_u(\partial)u = 0$. [proved] (C3-T6)

Theorem 3.3 Spectral form

$\sigma_p(u) = V(J_u)$ is finite; $\mathcal{R}u \cap \mathrm{EP} \cong \mathcal{R}/J_u$; $\mathrm{Ann}(u) = (P) \iff \sigma_p(u) = \varnothing$ and $G_u \sim P$. [proved] (P4-T2)

The condition $\sigma_p = \varnothing$ alone is not sufficient

$u = e^{x^2}$ solves $\tau^2 u = 0$ and has $\mathrm{Ann}(u) = (\tau)$ principal and no eigenvectors, but it does not identify τ^2: $G_u = \tau \not\sim \tau^2$.

3.3 How much data is enough?

Data model	Result	Status
Finite sum of exponentials $\sum_{k\le M} q_k e^{\langle z_k,\cdot\rangle}$	$\mathrm{Ann}(u)$ computed from $Q(z_k+\partial)q_k = 0$; Bézout threshold: more than $n \deg P_i$ points on each factor $\Rightarrow$ identified up to degree n; threshold not improvable uniformly	[proved] (C3-T2)
Jet of u at one point, $u \in \mathrm{EP}$, $d = \dim \mathcal{R}u$	Jet of order $2d-1$ determines $\mathrm{Ann}(u)$; optimal already in one variable	[proved] (C3-T3)
Evolution $u_t = a(\partial_x)u$	Identified iff $u \notin \mathrm{EP}$; $\tau - b \in \mathrm{Ann}(u)$ pins a in degree $\le n$ iff $\sum(\deg_x q_k + 1) \ge n+1$	[proved] (C3-T4)
Wave, Klein–Gordon, telegraph, beam	Wave: both profiles f, g non-EP; the others irreducible: $u \notin \mathrm{EP}$	[proved] (C3-T5)
Symmetry: $\ell(\partial)u = 0$ with ℓ linear	Then $\mathrm{Ann}(u) \ne (P)$ unless $P \sim \ell$: a translation symmetry destroys identification	[proof written, not audited] (C3-P3)
Restriction to a subdomain	Identifiability can be lost on a smaller Ω' (transport with a compactly supported bump); regained for elliptic P by analyticity	[proof written, not audited]/[from cited source] (C3-P5)

3.4 What no finite data can show

Theorem 3.4 No finite certificate

For every K and every $u \in \ker P$ there is an exponential-polynomial $v \in \ker P$ with the same K-jet at p_0 and $\mathrm{Ann}(v) \supsetneq (P)$. On convex Ω the same holds for any finitely many continuous linear functionals (Malgrange); on a non-convex set a single functional can certify. [proved] (jets) / [from cited source] (functionals) (C3-P1)

Theorem 3.5 Black-box impossibility, decidability with a bound

Model: an algorithm sees jets of u and zero tests $Q(\partial)u \stackrel{?}{=} 0$. (i) No algorithm decides "Ann$(u) = (P)$" correctly for all u without extra data. [proved] (P4-T5) (ii) Given an integer $B \geq \dim(\mathcal{R}u \cap \mathrm{EP})$ it is decidable, with test degree $B + \deg P - 1$. [proved] (P4-T4)

Theorem 3.6 A decidable class

For analytic solutions with D-finite Cauchy data the question is decidable in the holonomic model, modulo a computational assumption (AC) about algebraic numbers. [proved mod. (AC)] (P4-T6, OP-ALG-T1)

Takeaway. Identification is an infinite statement about u. It cannot be verified from finitely many numbers, but one extra number (a bound on how much of u is exponential-polynomial) makes it a finite computation. Whether that bound is the least possible is open ([open] P4-Q-min).

3.5 Variable coefficients: first limits

If coefficients depend on x the ring of operators becomes the Weyl algebra A_1 (polynomial coefficients), which is simple. Consequences:

- the restriction of the algebra to the module $A \cdot u$ is always injective and never identifies (the example $u = e^x$, dictionary $\{1, \partial\}$) [proved] (P4-C1);
- for continuous coefficients, identification of order-≥ 2 ODEs is impossible [from cited source] (Scholl et al.);
- for P monic in ∂_t with polynomial coefficients and non-similar factors there is an exact criterion, and a negative theorem in the algebra A_2 [proved] (OP-ALG-T4, OPR2-T3). Chapter 8 develops the structured cases fully.

3.6 What is not proved

- The exact condition for a *fixed* P on Ω with holes is Chapter 4.
- Minimality of the bound B [open].
- Identification of variable-coefficient PDEs in the Weyl algebra A_2 in general: only the exact criterion for monic P and the negative theorem (above).

CHAPTER 4

Decomposition

In this chapter

Starting from	$P = c\prod_{j=1}^{s} Q_j$ with $Q_j = P_j^{m_j}$ pairwise coprime, $u \in \ker P$ on an open $\Omega \subset \mathbb{R}^2$. Proofs: Volume II, Chapters 4 and 5.
Why this chapter	A solution of a composite equation is a sum of solutions of the simpler pieces. We say precisely which pieces, how unique they are, what topology on Ω is needed, and when the pieces keep the Sobolev regularity of u.
Role in the volume	Second link of the chain. The factor pieces are the "components"; the next chapter asks what is lost when one splits further, into additive terms.



4.1 Existence, ambiguity, and the topology of Ω

Theorem 4.1 Decomposition by factors

On a connected open $\Omega \subset \mathbb{R}^2$ with $H^1_{dR}(\Omega) = 0$ (equivalently simply connected: rectangles, convex sets, a slit disc), in $\mathcal{D}'$ and in C^∞:

(i) $\ker P = \sum_j \ker Q_j$.

(ii) The ambiguity of the splitting has dimension exactly $\delta(P) = \sum_{i<j} \dim \mathcal{R}/(Q_i, Q_j) = \sum_{i<j} m_i m_j \sum_z I_z(P_i, P_j)$, and consists of exponential-polynomials.

(iii) Unique $\iff$ $V(Q_i) \cap V(Q_j) = \emptyset$ for all $i \neq j$.

[proved] (C12-T1, T1′)

Operator	Factors	$\delta(P)$	Ambiguity
Wave $\tau^2 - \xi^2$	$\tau \mp \xi$	1	constants
Beam $\tau^2 + \xi^4$	$\tau \mp i\xi^2$	2	span$\{1, x\}$
Heat$\times$transport $(\tau - \xi^2)(\tau - \xi)$	2	2	span$\{1, e^{t-x}\}$
Parallel transports $(\tau - \xi)(\tau - \xi - 1)$	2	0	none (unique)
$\tau\xi$	τ, ξ	1	constants

Intuition

Two factors share "zeros" where their curves $V(Q_i), V(Q_j)$ cross; at a crossing point the exponential $e^{\langle z, \cdot \rangle}$ belongs to *both* kernels, and you can shift it from one component to the other freely. The number of such free shifts, counted with intersection multiplicity, is $\delta(P)$.

Theorem 4.2 Holes break it

If Ω is connected but not simply connected, there is P (the Laplacian) and $u \in C^\infty$ with $P(\partial)u = 0$ that has no splitting into $\ker(\tau - i\xi) + \ker(\tau + i\xi)$: take $u = \log|(t, x) - a|^2$

around a hole α. For a *fixed* P the exact condition is cohomological: P and $\mathcal{A}$ decide (for $\tau\xi$ in C^∞ on the punctured disc it still splits; in $\mathcal{D}'$ it does not). [proved] (C12-P10, OP-FUN Q6)

4.2 Explicit projectors

For an evolution factor $\prod_j(\tau - a_j(\xi))$ with a_j distinct functions, the Lagrange formula gives the splitting by Fourier multipliers: the component of u at factor i is the multiplier $\prod_{j\neq i} \dfrac{\partial_t - a_j(\partial_x)}{a_i(\partial_x) - a_j(\partial_x)}$ applied to u, rational in ξ with poles exactly where two a_j meet. [proved] (II-T2) The projectors are polynomial iff the varieties are disjoint [proved] (II-T3); only factors admit one-value projectors [proved] (II-T4). The companion matrix is normal for $m = 2$ (earlier claims for $k = 2$ were wrong) [proved] (II-P5), and Wei–Norman holds globally iff certain Wronskians do not vanish [proved] (II-T7). The Vandermonde case (degree-one factors in τ) [proved] (C12-T2). A quadratic-in-t factor is reducible when its discriminant is a square: [proved] (Lean-checked) (C12-P2, P3).

4.3 Regularity: when the pieces stay in H^m

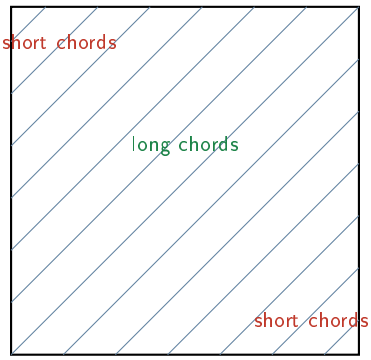


Figure 4.1: A square and the characteristics of $\partial_t + \partial_x$. Chords near the two corners are arbitrarily short. On a short chord the functions $e^{d_1 t}$ and $e^{d_2 t}$ become nearly collinear, the splitting is ill-conditioned, and the global L^2 splitting is lost (the counterexample C12-C1 (Volume II, Chapter 5)).

Statement	Meaning	Status
Decomposition does not preserve H^m: $u \in H^m$, components $\notin H^m$, loss of exactly one derivative	Parallel transports on the square.	[proved] (C12-C1)
Chord criterion: components in L^2 iff $\inf \ell > 0$ iff $c_F < 1$	Geometry of Ω against the characteristics decides.	[proved] (C12-T3)
H^m splitting for any N factors $\iff$ sum of kernels closed ($\gamma_N > 0$) and dense	Closedness and density are separate conditions.	[proved] (C12-P7, P2-T3, P2-C2)
Pairwise Friedrichs cosines do not decide for $N \geq 3$	Abstract counterexample in Hilbert space.	[proved] (abstract)
Density of $\sum \ker_{H^m} Q_j$ in $\ker_{H^m} P$ for every P, every bounded convex $\Omega \subset \mathbb{R}^2$, every m	On convex sets only $\gamma_N > 0$ remains to check.	[proved mod. Hörmander 10.2.1, Stein] (OPF5B-T1)
Cases where it holds: $\tau\xi$ on a rectangle; wave on a square	Regularity depends on geometry, not on P alone.	[proved] (C12-P4)
Density outside convex domains	Open.	[open]

4.4 Variable coefficients

For polynomial coefficients in x (ODE in x, evolution in t) the "sum of kernels" becomes the kernel of the *least common left multiple* $\mathrm{LCLM}(L_1, L_2)$, not of the product $L_1 L_2$. [proved] (C12-P5, R2-T4) This is where the Weyl algebra enters Volume I; Chapter 8 treats it.

> **Takeaway.** Existence of the factor splitting is purely algebraic once Ω is simply connected. Everything delicate happens at the *boundary* of Ω (shrinking chords) or in the *topology* of Ω (holes).

CHAPTER 5

Terms inside a factor, and the interaction term

In this chapter

Starting from	An evolution factor $\tau - a(\xi)$ with $a = \sum_{n=1}^{N} a_n$, the a_n polynomials in ξ (the terms: advection, diffusion, dispersion…); $u \in \ker$ on a bounded rectangle; base time t_0. Proofs: Volume II, Chapter 6.
Why this chapter	Chapter 4 splits a solution per factor. Inside one factor the natural question of the original idea is per *term*: how much of u comes from advection alone, from diffusion alone, and how much from their interplay? The answer is partly positive (an exact formula) and partly negative (an instability).
Role in the volume	Third link of the chain. Defines E_{int}, computes it, prices it, and shows how robust it is.



5.1 Definition and exact formula

Each term evolves the initial datum alone: $u_n(t) = e^{(t-t_0)a_n(\partial_x)}u_0$. The interaction term is what is missing:

$$E_{\text{int}}(t) = u(t) - u_0 - \sum_{n=1}^{N} \big(u_n(t) - u_0\big).$$

Theorem 5.1 Closed form for commuting terms

For u_0 of exponential type (exponential-polynomials, or limits of them in the class $\mathfrak{B}$),

$$E_{\text{int}}(t) = \sum_{|S|\geq 2} \prod_{n\in S} \big(e^{(t-t_0)a_n(\partial_x)} - \mathrm{Id}\big)u_0, \qquad E_{\text{int}}^{\text{mult}} \equiv 0.$$

For $u_0 = \sum c_k e^{\zeta_k x}$ this is $\sum_k c_k\, e_{\geq 2}(z(\zeta_k))e^{\zeta_k x}$ with $z_n = e^{(t-t_0)a_n(\zeta_k)} - 1$ and $e_{\geq 2}(z) = \prod(1+z_n) - 1 - \sum z_n$. [proved] (P1-T2)

For two terms, $E_{\text{int}} = z_1 z_2\, u$ in each mode: the interaction is the *product* of the two single-term responses. For advection–diffusion at a stationary state it is $-4\sinh^2(\theta/2)\,u$ with $\theta = c^2(t-t_0)/D$, so equilibrium is pure interaction.

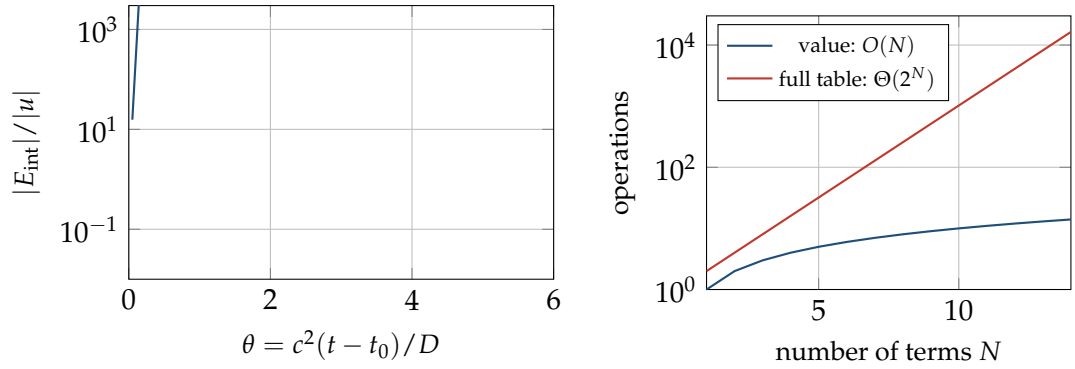


Figure 5.1: Left: relative size of E_{int} for advection–diffusion at a stationary state; it is $O(\theta^2)$ for small θ and grows exponentially. Right: asking for the interaction at one point costs linear time; listing every sub-interaction c_S costs exponential time.

5.2 Cost, and an instability

Question	Answer	Status
Value of E_{int} at a point	$O(M(N + \sum d_n))$ for M simple exponentials	[proved] (P1-T3)
Aggregates by order $\lvert S\rvert = k$	$O(MN^2(L+1)^2)$	[proved] (P1-T3)
Full table of c_S	$\Theta(M2^N(L+1)^2)$; "$O(e^n)$" holds only for the full table	[proved] (P1-T3)
Continuity of $u \mapsto E_{\text{int}}$ in $H^m(\Omega)$	*False* for advection–diffusion, every m, every t_0: the diffusion flow alone amplifies real modes by $e^{D\zeta^2 t}$ while the true solution is damped	[proved] (P1-P1(e))
Without a function class, are single-term flows determined by u?	No: transport has a free extension, the heat equation has Tychonoff solutions with zero datum	[proved] (P1-P1(a)–(d))
Exception for proportional terms	E_{int} continuous in H^m only for proportional terms; discontinuous for ≥ 2 non-proportional terms	[proved] (P1-Q6, OP-ANA)

Consequence for any stable algorithm

No regularisation can remove $E_{\text{int}}^{\perp}$ (the part of the interaction outside the span of the dictionary); ℓ^1 penalties act only on the finite ambiguity; no linear encoder suffices (the log-symbol is needed); and no uniformly Lipschitz regularisation of the unbounded map exists. [proved] (P2-T1..T5, P2-C1)

5.3 Time-dependent perturbation, boundaries, non-commuting terms

1. **Dyson–Phillips** [proved] (P3-T1..T6). For an even-order dissipative operator plus a lower-order L^∞ perturbation (also with Dirichlet conditions) the Dyson series converges for every t (Miyadera–Voigt, Engel–Nagel). The interaction from the *boundary* is $\Theta(t^{k/2+3/4})$ where k is the order of contact of the datum with the boundary

[proved] (P3-T3, OP-F5A); for compatible data the norm of the boundary-induced E^{mult} is $|c|\,(\tilde{N}_k/k!)\,D^{k/2-1/4}\sqrt{u_0^{(k)}(0)^2+u_0^{(k)}(L)^2}\;t^{k/2+3/4}\,(1+O(t^{1/2}))$ for every $k \geq 1$.

2. **Lie-algebra classification** (terms that do not commute).
   - Finite solvable Lie algebra generated by the terms: exact E^{mult} with $d = \dim \mathfrak{g}$ entire functions [proved] (P1-T4) (Wei–Norman; example $\partial_x^2 + x$, Ornstein–Uhlenbeck).
   - Finite non-solvable (e.g. $\mathfrak{sl}_2$): closed form only locally; globality can fail [proved] (P1-P5).
   - Infinite (e.g. $\partial_x^2 + x^k$, $k \geq 3$): no finite product of term flows reproduces $e^{t(X+\varepsilon Y)}$ [proved] (P1-T5); series only.
3. **Boundary conditions** on an interval break commutativity: the boundary creates an E^{mult} of first order [proved] (P1-P6).
4. **Repeated factors, non-monic**: multiplicities give $E^{\mathrm{mult}} = 0$; degree ≥ 2 in τ gives $E^{\mathrm{mult}} \not\equiv 0$; non-monic P loses uniqueness of the splitting [proved] (P1-Q2, OP-FUN).
5. **Second-order evolutions** (wave, Klein–Gordon): the companion blocks do not commute, so the first-order calculus fails; Gauss form mode by mode [proved] (II-T7, P1-P4).

The one continuous per-term splitting found

A balance decomposition of u (a conservation-law split) [proved] (P3-T6) is continuous but needs an external selection principle; the Faddeev/Lippmann–Schwinger route (RT-S2a) is parked.

Takeaway. Per-factor splitting is exact and stable on suitable domains; per-term splitting is exact only inside a function class of exponential type and is intrinsically unstable in Sobolev spaces. That instability is not a defect of the formula: it is a property of the question.

CHAPTER 6

Nonlinear equations

In this chapter

Starting from	Scalar nonlinear evolution equations: viscous Burgers $u_t + uu_x = \nu u_{xx}$, KdV, KdV–Burgers, Fisher–KPP; smooth data u_0; the term-by-term question of Chapter 5.
Why this chapter	For a nonlinear equation the single-term flows do not commute and superposition fails, so "component per term" needs a new definition, and identification needs a new frame (a dictionary). We give both and compute them on the models where everything is explicit.
Role in the volume	Fourth link; it re-uses Chapters 3–5 through the jet expansion and the annihilator machinery. Proofs of the core are in Volume II, Chapter 7.



6.1 The nonlinear interaction term (A9)

Let the equation be $u_t = \sum_n T_n(u)$ with each T_n a nonlinear operator (e.g. $T_1 = -uu_x$, $T_2 = \nu u_{xx}$) and S_n^t the flow of $u_t = T_n(u)$. The A9 interaction term is

$$E_{\mathrm{int}}^{\mathrm{A9}}(t) = u(t) - u_0 - \sum_n \left(S_n^t u_0 - u_0\right).$$

Theorem 6.1 Jet in time

The coefficient of t^2 of $E_{\mathrm{int}}^{\mathrm{A9}}$ is $\frac{1}{2}\sum_{n\neq m} T_n'[u_0]\,T_m(u_0)$ (the commutator-type interplay). For Burgers it equals $-\nu\,(u_0u_0''' + 2u_0'u_0'')\,t^2$. [proved] (S4T-T1)

Theorem 6.2 Closed form before the shock

Before the shock time t^* the interaction has a closed form to first order in ν with remainder $O(\nu^2\varepsilon^2t^3)$, and it blows up at t^*. [proved] (S4T-T2, S4T-P3)

6.2 Across the shock

Let t^* be the first shock time and $\nu \to 0$.

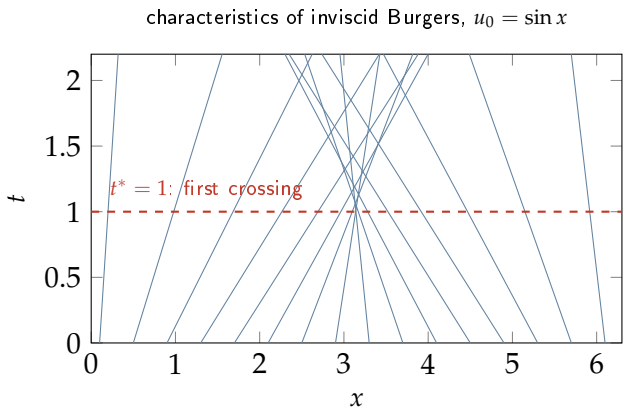


Figure 6.1: For $u_0 = \sin x$ the characteristics $x = x_0 + t \sin x_0$ first meet at $t^* = 1$, where the inviscid solution becomes discontinuous. Before t^* everything is smooth and the closed form holds; after it a shock forms and the interaction term acquires size $\Theta(\nu)$ in L^1.

Statement	Meaning	Status
$\|E_{\text{int}}^{\text{A9}}\|_{L^1} = \Theta(\nu)$ at fixed $t > t^*$, under nondegeneracy (H_t) (holds e.g. for $\varepsilon \sin x$); each shock contributes $4\nu \log(2\cosh(\rho/2))$	Positive answer to the term question in L^1 even after the shock.	[proved mod. [F1]] (OPA-T1)
Uniformly in t: $2\nu \log \frac{1}{\max(\|t-t^*\|,\sqrt{\nu})} + O(\nu)$, so $\sup_{t \le T} = \nu \|\log \nu\| + O(\nu)$, coefficient exactly 1, explicit constant Γ at the shock time (Pearcey constant $P = 1.188654$; $\Gamma(\sin) = 2.57495$)	Exact uniform rate.	[proved mod. [F1]] (OPF5A-T1/T2)
General case with several shock births	Reduces to a localisation lemma (L-loc).	[conditional]
Splitting error of the AB splitting (advection then diffusion) $\Theta(\sqrt{\nu})$ in L^1	The splitting error is worse than the interaction itself.	[proved] (OPF5D-T6)
Jumps of E^{add} in L^∞ / H^s independent of ν	Forced by the A9 definition: the flow of uu_x alone forms a shock.	[proved mod. [F1]] (S4T-T3)

[F1] is the uniqueness of entropy solutions, reduced to Kružkov (1970, Thm 1 §3), checked on the primary text.

6.3 Identification relative to a dictionary

In the nonlinear case no ring of operators is given. One fixes a *dictionary* V of candidate terms (e.g. $\{u, u_x, uu_x, u_{xx}, u_{xxx}, \dots\}$) and asks which linear combinations vanish along u.

Theorem 6.3 Dictionary identification

There is an exact reduction $0 \to [F] \cap V \to I(u) \cap V \to J_u \cap \psi(V) \to 0$ (valid for u merely C^∞ on any Ω): u identifies its equation in V *modulo the differential consequences* of the equation F. In complete dictionaries the literal statement fails except for $(2, n)$. [proved] (S4ID-T1, P1)

Equation	Criterion	Status
Burgers, N shocks	Identifies in the evolutionary dictionary $\iff d \le N$ (d = number of unknown terms)	[proved] (S4ID-T2, OP-T7)
KdV, N solitons	The annihilator is principal, generated by Π_N of total degree $(N+2)2^{N-1}$; identifies $\iff d < (N+2)2^{N-1}$	[proved mod. Novikov 1974] (S4ID-T3, OPF5C-T2)
Exact certificate $\deg \Pi_2 = 8$ (generic)		[proved]
Closed form $\Rightarrow$ not identifying	A closed-form u lives on a smaller variety	[proved] (ALG-P3, S4ID-T4)
Variable coefficients, monic in ∂_t, non-similar factors	Exact criterion	[proved] (OP-ALG-T4, OPR2-T3)

6.4 Integrable models, scattering, analyticity

- **Hirota N-solitons.** The interaction is pairwise and localised: E_{int} is determined by pairs, integrates to 0, and costs $O(N^3)$ [proved] (S4-INTEG, Hirota).
- **Transport lemmas.** Analyticity radius of the solution of Burgers with datum U_0: at least $2\pi\nu/\operatorname{osc}(U_0)$, optimal [proved] (F4L-P3). (The "infinite radius" claim of a cited preprint is not proved in that text; its own equation has denominator zeros at finite δ for every x [proved] (KIEFFER-C).)
- **Scattering identification.** KdV with no reflection: Novikov's centraliser characterises the potential [proved] (INT-T1(C), OPF5D). NLS with multiplicity: identification by the scattering data; the version without multiplicity is [false]. Lax and zero-curvature relations hold for all n (a sign error at $n = 0$ was found and corrected by the reviewer) [proved] (OP-P17/P18). Open: for analytic data with nonzero reflection is there an autonomous ODE in x? ([open] INT-Q1: three proved obstructions, among them the cubic relation of Airy).
- **KdV–Burgers.** The jet is proved; the dispersive-correction remainder is $O((\nu + |d|)^2)$ before the shock [proved mod. local well-posedness] (OPR2-T5).
- **Fisher–KPP.** A jet-level statement with non-uniform-in-t estimate [proved] (S4T-P5). No sharp asymptotic analogue of Burgers is claimed.
- **Exact numbers.** For the minimal syzygy degrees $\delta_N(K)$ nine exact values were obtained by computer-assisted calculation and independently reproduced; bounds for all N [conditional]; $\delta_N(2N-1) = N+2$, $\delta_N(2N-2) = N+4$, $\delta_N(2N) = N+1$ are conjectures for general N ([open] OPF5C-Q1).

Takeaway. Nonlinearity changes the *object* (single-term flows with a shock) but not the *question*. Exactly where the formula is closed (before the shock, solitons) the answer is explicit; across a shock the interaction does not vanish with the viscosity: it shrinks like $\nu|\log\nu|$.

CHAPTER 7

Systems of equations and gauge

In this chapter

Starting from	A *system* $P(\partial)u = 0$ with $P \in \mathcal{R}^{l\times k}$ a matrix of constant-coefficient operators, $u \in (C^\infty(\Omega))^k$ or $(\mathcal{D}'(\Omega))^k$, $\Omega \subseteq \mathbb{R}^n$ open connected, $\mathcal{R} = \mathbb{C}[\zeta_1, \dots, \zeta_n]$. Origin: the user's Problem 2 (matrix coefficients, e.g. the 4×4 Dirac operator).
Why this chapter	Chapter 3 identified a scalar operator. For systems the ring is the same but the object that carries the equation is a *module* $M_P = \mathcal{R}^{1\times k}/\mathcal{R}^{1\times l}P$, and the same questions need module-theoretic answers: when does one solution determine the system, and what ambiguity (a change of basis, a gauge) is unavoidable?
Role in the volume	Fifth link. It generalises Chapters 3–5 to matrices; its results feed the dictionary (Chapter 10, class U4). Status: Phase 8, independently audited with fixes ([proved, F8]); registration in the ledger is pending.



7.1 When does one solution identify a system?

Solutions of P correspond to module maps $M_P \to \mathcal{A}$. A solution u *identifies* P when its annihilator (the row-vectors r with $r(\partial)u = 0$) equals the row module of P, i.e. the induced map $\varphi_u : M_P \to \mathcal{A}$ is injective. For a maximal ideal $\mathfrak{m}_\lambda$ the *socle* $\operatorname{soc}_{\mathfrak{m}_\lambda}(M)$ is the space of elements killed by $\mathfrak{m}_\lambda$.

Theorem 7.1 Existence criterion

Let M be a finitely generated $\mathcal{R}$-module and Ω connected nonempty. An injective map $M \hookrightarrow \mathcal{A}$ exists iff $\dim_{\mathbb{C}} \operatorname{soc}_{\mathfrak{m}}(M) \le 1$ for every maximal ideal $\mathfrak{m}$. For a system: some solution identifies P iff $\dim \operatorname{Hom}_{\mathcal{R}}(\mathcal{R}/\mathfrak{m}, M_P) \le 1$ for all $\mathfrak{m}$; then it can be taken C^∞. [proved, F8] (SYS-EX)

Intuition

A joint eigenvector for the eigenvalue λ has to be sent to a multiple of $e^{\lambda\cdot x}$, and on a connected domain that space is one-dimensional. So two independent eigenvectors at the same λ can never be told apart by any single function: a "socle obstruction". The theorem says this is the only obstruction.

Statement	Meaning	Status
Socle of dimension ≥ 2 at some $\mathfrak{m} \Rightarrow$ no identifying u (SYS-N1)	Obstruction.	[proved, F8]
ODE systems: identifying u exists iff the invariant factors satisfy $e_1 = \cdots = e_{r-1} = 1$	Smith normal form.	[proved, F8] (SYS-T0)
Square system, $\det P \neq 0$, $n \geq 2$ variables: an identifying u always exists on every connected Ω; associated primes are the prime factors of $\det P$	Depth argument: $\operatorname{grade}(\mathfrak{m}, \mathcal{R}) = n \geq 2$ kills the socle.	[proved, F8] (SYS-C1)
Exponential-polynomial data: M_u has finite length (difference with prior work of Schindelar–Levandovskyy–Zerz)	Chapter 3's EP part, for matrices.	[proved, F8] (SYS-R1)
Lamé operator, $n = 3$: associated primes $\{q_P, q_S\}$, multiplicities $(1, 2)$; criterion by div/rot of the field; needs $\lambda + 2\mu \neq 0$	Elasticity: pressure and shear waves.	[proved, F8] (LAME-3)

7.2 Sum of kernels for systems

Theorem 7.2 Kernel of a sum, systems version

$\ker P_1 + \ker P_2 \subseteq \ker \mathrm{LCLM}$ holds on every Ω; equality holds when $\mathrm{Ext}^1(N, \mathcal{A}) = 0$, e.g. for N of finite length with $H^1_{dR}(\Omega) = 0$. The ambiguity is $\delta_{\mathrm{sys}} = \ell(N)$. [proved, F8] (SYS-S1)

The converse is false in general

On a simply connected but non-convex Ω with N of infinite length, the converse fails, in $C^\infty(\Omega)$ and in $\mathcal{D}'(\Omega)$ [false] (SYS-S1-C).

7.3 Interaction for systems, and the Dirac case

- **Dirac in 1+1 dimensions.** The condition (H) of the commuting/Lie theory holds, so the impossibility theorem for infinite adjoint orbits applies [proved, F8] (SYS-E1).
- **The Lie algebra is a twisted current algebra:** $\mathrm{Lie}\langle X, Y\rangle = (\mathfrak{sl}_2 \otimes \mathbb{C}[\xi])^\theta$ [proved, F8] (SYS-E2).
- **Euler ZXZ formula explicit**; the time dependence is not polynomial and the multipliers are nonlocal [proved, F8] (SYS-E3).
- **Dirac in Hamiltonian form.** Let $\mathcal{C}$ be the class $L = I_4\partial_t + \sum_{k=1}^3 \alpha^k \partial_k + B$ with arbitrary 4×4 complex α^k, B (no Clifford relations assumed). If $\psi \in \ker L_0$ identifies L_0 and $L \in \mathcal{C}$ with $L\psi = 0$, then $L = L_0$: the matrices α^k, B are determined exactly, with no left GL ambiguity and no conjugation. [proved, F8] (H-DIRAC-1)

Idea of the proof. The rows of L lie in $\mathrm{Ann}(\psi) = \mathcal{R}^{1\times 4} L_0$, so $L = V L_0$. The difference $(V - I)L_0$ has degree ≤ 1 and no ∂_t. Its top-degree part is $W_d(\tau I + \alpha_0 \cdot \xi)$, and $\det(\tau I + \alpha_0 \cdot \xi) = \tau^4 + \ldots \neq 0$ makes this matrix a non-zero-divisor; so $W = 0$. (Proof in Volume II, Chapter 8.)

Existence holds for every $L_0 \in \mathcal{C}$ (the determinant is monic in τ, so Theorem above with $n = 4$ applies). [proved, F8] (H-DIRAC-2) The criterion in 1+1 dimensions: $\mu = 2$, equivalently $\psi_2 \notin \mathrm{EP}$.

7.4 Gauge

Two sets of coefficients related by a gauge transformation give the same physics and a transformed solution. Identification can therefore only be hoped for *up to gauge*.

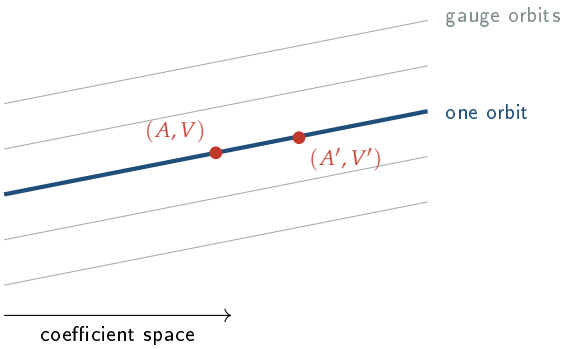


Figure 7.1: Coefficients are only meaningful modulo gauge: each oblique line is an orbit. Data invariant under gauge can at best determine the orbit; the quotient is described by the field strength F and the holonomies.

Statement	Meaning	Status
Covariance: identification commutes with gauge transformations (G-L0)	Basis-independent.	[proved, F8]
Gauge-invariant data determine only the orbit; the quotient is $(F, \text{holonomies})$ (G-T2)	Aharonov–Bohm: holonomies are invisible to F.	[proved, F8], [from cited source]
Magnetic Schrödinger $iu_t = -(\nabla - iA)^2 u + Vu$, $n \geq 2$, $u \neq 0$ everywhere: one solution does not identify (A, V) even up to gauge, even non-stationary; the set of compatible pairs is an infinite-dimensional affine space $(A, V) + \{(-k/\rho^2, (\lvert j\rvert^2 - \lvert j + k\rvert^2)/\rho^4) : \operatorname{div} k = 0\}$	A divergence-free current can be added without changing u.	[proved, F8] (SYS-N3)
Fixed gauge does not identify Maxwell	Remark only.	remark (G-T0)

7.5 Open and false

- Statement "crossing of eigenvalue branches $\Rightarrow$ pole of the projector" for a derogatory matrix is [false] (counterexample $\operatorname{diag}(\xi, -\xi)$); the corrected version is [open] (II-T2(b′)).
- Finite products of flows with polynomial-in-ξ times: [open] (Q-S8SYS-1).

Takeaway. For systems the only obstruction to identification is a repeated eigen-direction (a socle of dimension ≥ 2); square nonsingular systems in at least two variables never have it. What a solution cannot give is the gauge: it determines coefficients only up to the orbit, and in the magnetic case not even that.

CHAPTER 8

Variable coefficients: identifying a potential

In this chapter

Starting from	Schrödinger-type equations $iu_t = -u_{xx} + Vu$ with an unknown real potential $V = V(x)$ or $V(t,x)$ (bounded, locally L^∞), ODEs $L_u y = 0$ with polynomial or rational coefficients, and non-autonomous evolutions. Origin: the user's Problem 1 (varying V, e.g. Schrödinger, Dirac).
Why this chapter	With variable coefficients the ring of operators is no longer commutative (it is a Weyl algebra), Chapter 3's criterion does not apply, and we first restrict to *structured* classes where the question has a precise answer: the potential itself (B1: identify V).
Role in the volume	Sixth link. Chapter 9 treats the other goal (B2: remove V); Chapter 10 uses the rank theorem proved here. Status: Phase 8, independently audited with fixes.



8.1 Identifying V itself

For $iu_t = -u_{xx} + Vu$ with $u \not\equiv 0$ one solves for $V = (iu_t + u_{xx})/u$ wherever $u \neq 0$. So V is determined by u off the zero set; the question is what happens on it.

Theorem 8.1 Potential $V = V(x)$

Let $V \in L^\infty_{\mathrm{loc}}$. For $u \not\equiv 0$ the zero set X_0 of $u(t_0, \cdot)$ in x is discrete, and every non-zero solution identifies V almost everywhere. Proof uses lateral unique continuation for $i\partial_t + \partial_x^2$ (Carleman estimate with weight $-s + \mu s^2 - \kappa(t - t_0)^2$). [proved, F8] (VAR-T1(b))

Theorem 8.2 Potential $V = V(t,x)$: silent bands

$V(t,x)$ is identified by u iff the interior of the zero set $Z(u)$ is empty (resp. $|Z| = 0$). The naive statement "every $u \not\equiv 0$ identifies $V(t,x)$" is [false] (Tychonoff-type solutions vanish on a band where V is free). [proved, F8] (VAR-T1(c))

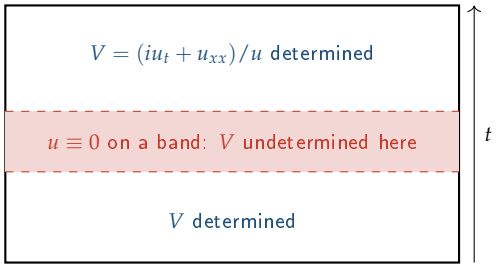


Figure 8.1: Where u vanishes on a band the equation says nothing about V. The band-free case is exactly the identifiable one.

Statement	Meaning	Status
Rank-one annihilators for large ∂-closed classes with $u \neq 0$	Structure of Ann over a differential field.	[proved, F8] (VAR-N1)
Affine families: compatible parameters $= \theta_0 + \ker \Phi_u$	Identification = injectivity of a linear map Φ_u.	[proved, F8] (VAR-T1(a))
Stability: Lipschitz $C^{1,2} \to$ potential, not continuous from C^1, Hölder $1/3$ sharp	Quantitative well-posedness.	[proved, F8] (VAR-P3)
Gauge-fixed form: V determined only modulo constants; redundancy of (P, q)	Constant phase e^{-ict}.	[proved, F8] (VAR-P4)
Dirac 1+1 with potential: $V = \langle \psi, r \rangle / \lvert\psi\rvert^2$, matrix case		[proved, F8] (VAR-P5)

8.2 ODEs: structure and Galois theory

Theorem 8.3 Linear ODEs

For $L_u y = 0$: $\mathrm{Ann}(u) = \mathcal{D} L_u$ (left ideal in the Ore algebra), the order of the minimal equation is $\dim \mathrm{span}(Gu)$ where G is the differential Galois group; irreducible $\Rightarrow$ every non-zero solution identifies; existence of an identifying solution $\iff$ the module is cyclic ($m_i \leq \dim S_i$). [proved, F8] (Galois part [from cited source]: $E^G = K$) (VAR-T2a, VAR-T6)

- Airy is irreducible [proved, F8] (VAR-E1). The harmonic oscillator $-y'' + x^2 y = Ey$ is reducible iff $E \in \pm(2\mathbb{N} + 1)$; exact identification [proved, F8] (VAR-E2).
- Prior work: Schindelar–Levandovskyy–Zerz (2011) [from cited source], abstract only; the new part is the restriction to hyperexponential solutions.

8.3 Dynamical Lie algebras, Pöschl–Teller, non-autonomous

Statement	Meaning	Status
The dynamical Lie algebra $\mathfrak{g}_V$ is finite-dimensional iff $V \in \operatorname{span}\{1, x, x^2\}$	Only quadratic potentials give a finite Lie algebra.	[proved, F8] (VAR-P8(a))
Wei–Norman exponentiation is finite iff V is quadratic		[proved, F8] (VAR-P8(b))
Pöschl–Teller with $\ell \in \mathbb{N}$: explicit U_ℓ via a Darboux (Crum) chain; explicit E_{int}	Reflectionless potentials.	[proved, F8] (VAR-P8(c,d))
Non-autonomous second order: E^{mult} is the two-time commutator $+O(t^3)$; commuting at equal times is not enough		[proved, F8] (VAR-P9)
Dyson–Phillips: existence, uniqueness, evolution property, unitarity		[proved, F8]/[from cited source] (VAR-P11)
Stark: global exponentiation; oscillator: the product of three factors is valid iff $\cos t \neq 0$ (incl. lens $t = \pi/4$); bounded Magnus (Casas)		[proved, F8]/[from cited source] (VAR-T10)
Floquet: bounded logarithm always; differential logarithm for $V = v(t)$ unbounded	General case open.	[proved, F8]/[open] (VAR-P12)
Weyl algebra A_2: L monic in ∂_t identifies iff B-linear independence; $m = 1 \iff u$ not D-finite; existence for free Schrödinger; $x\partial_t$ never identified	Answers Chapter 3's open question C3-Q2 in the monic case.	[proved, F8] (VAR-C3Q2)

Takeaway. The potential is identified wherever the solution is nonzero; the only failures are silent zones. The exact algebraic criteria (Galois, cyclic modules, Lie algebras) say which *families* of potentials admit closed-form evolution: essentially the quadratic ones and the reflectionless Pöschl–Teller family.

CHAPTER 9

The analytic encoder: removing a potential

In this chapter

Starting from	$L_q = \partial_x^2 - q(x)$ on an interval J with q known (or the time-dependent $i\partial_t - H_0 - V(t)$), compared with the free operator $L_0 = \partial_x^2$. No learning: only explicit transformations.
Why this chapter	Chapter 8 identified V. The second goal for variable coefficients (B2) is to treat V as *noise*: to find a transformation W that turns solutions of the equation with potential into solutions of the free equation, so that the machinery of Chapters 3–5 (constant coefficients) applies to Wu.
Role in the volume	Seventh link. It is the bridge from Chapter 8 back to the constant-coefficient theory and is used by the dictionary chapter (rule R6, change of frame). Status: Phase 8, audited with fixes (rev. 2).

Identify → Decompose → Terms → Nonlinear → Systems → **Variable coeff.** → Dictionary + limits → Algorithm

9.1 What is an encoder?

Definition 9.1 Structured encoder (ENC-D)

A linear operator W on a function space $\mathcal{Y}$ is a *structured encoder* for L_q if (E1) $\dim\ker W < \infty$ on $\mathcal{Y}$ itself (not only on solutions), (E2′) $WL_q = (L_0 + c)W$ for some constant c ("modulo constants": equality $c = 0$ is too strict), (E3) W is explicit: differential, pointwise plus gauge, or Volterra with a kernel given by a well-posed problem.

The condition (E1) is what makes the notion non-empty: with "injective on solutions" only, restriction to a region where the potential is autonomous would qualify. For constant $q \equiv c \neq 0$ there is no *strict* differential intertwiner of any order (comparing coefficients forces a polynomial of growing degree that violates the last equation) [proved, F8] (ENC-K0); with (E2′) $W = \mathrm{Id}$ works.

9.2 The Volterra transformation operator

Theorem 9.2 Symmetric Volterra operator (ENC-1)

Let $J = [x_0 - a, x_0 + a]$ be symmetric about x_0 (translate to 0), $q \in C^1(J;\mathbb{C})$ (complex allowed), $\Delta = \{(x,s) : |s| \le |x| \le a\}$. Then:

(1) There is a unique $K \in C(\Delta)$, C^2 on each half, with $K_{xx} - K_{ss} = q(x)K$ on Δ, $K(x,x) = \frac{1}{2}\int_0^x q$, $K(x,-x) = 0$.

(2) $Tf(x) = f(x) + \int_{-x}^{x} K(x,s)f(s)\,ds$ maps $C^2(J)$ to itself and for every $f \in C^2(J)$ (no boundary conditions): $L_q Tf = TL_0 f$, $(Tf)(0) = f(0)$, $(Tf)'(0) = f'(0)$.

(3) T is bijective on $C(J)$ and $C^2(J)$; $W = T^{-1} = \mathrm{Id} + \int_{-x}^{x} H(x,s)\cdot ds$ with H solving the same problem with $q \to -q$ in the equation and $H(x,x) = -\frac{1}{2}\int_0^x q$. $WL_q = L_0 W$.

(4) If $q \in C^\infty$, T, W are bijections of $C^\infty(J)$.

(5) $\|K\|_\infty \le \frac{aQ}{2} e^{a\sqrt{Q}}$ with $Q = \|q\|_\infty$; $\|K_q - K_{q'}\|_\infty \le C\|q - q'\|_\infty$; $\|T\|, \|W\| \le 1 + 2a\|K\|_\infty$ on $C(J)$ and $L^2(J)$.

(6) T depends on the base point x_0 already for constant $q \neq 0$.

[proved, F8] (written proof by Goursat iteration; [from cited source] in substance: Povzner 1948, Marchenko 1986, Levitan 1987). (ENC-1, ENC-U1, ENC-C1)

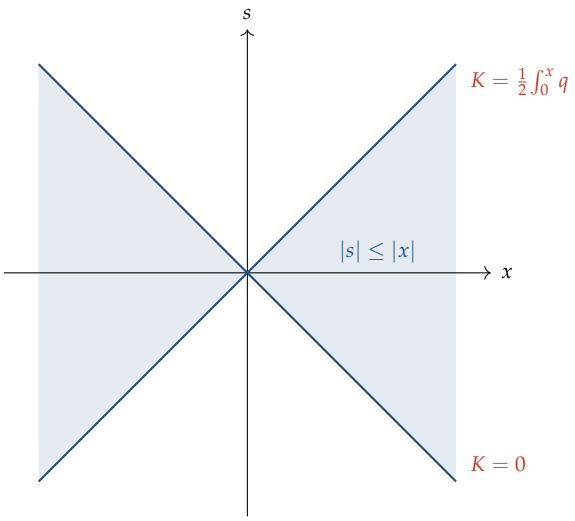


Figure 9.1: The domain Δ of the kernel $K(x,s)$: two triangles meeting at the origin. The kernel is determined by a characteristic (Goursat) problem with data on the two boundary lines $s = \pm x$; so $Tf(x)$ only uses f on $[-x,x]$ ("Volterra"), which is why no boundary conditions are needed.

Intuition

T is the identity plus a smoothing correction supported on the interval $[-x,x]$. It sends the free solution $\cos(kx)$ or $\sin(kx)/k$ to the solution of $L_q y = -k^2 y$ with the same initial value and slope at 0. In scattering theory this is the Gel'fand–Levitan–Marchenko kernel; here it is used purely as an explicit change of variables.

9.3 What cannot exist

Statement	Meaning	Status
Transport lemma at known q: W intertwines the Schrödinger flows (ENC-T2)	Not a tool for identifying V.	[proved, F8]
No time-independent encoder with finite kernel exists for a genuinely non-autonomous $V(t,\cdot)-c(t)$; encoders that exist depend on time, as interaction pictures $W(t)=U_0(\theta(t))W(0)U_V(t)^{-1}$; pointwise-plus-gauge representatives exist exactly for $V=\alpha(t)x^2+\beta(t)x+\gamma(t)$ (Bluman–Shtelen), $\theta\neq \mathrm{Id} \iff \alpha\not\equiv 0$	Even Avron–Herbst has $\theta=\mathrm{Id}$.	[proved, F8] (P0(a), (b)); [from cited source] (ENC-P0)
"The exceptions change the time"	First draft claim.	[false] (killed)
A *blind* encoder (one that works for every V in a class without knowing V) together with identifiability	The F6 claim.	[false]; reduction lemma proved (ENC-N1, ENC-N1′)
Poles obstruction		[proved, F8] (POLE-N)
K-coefficient sharper statement for $V\to 0$ only (D–G)		[from cited source], to check (ENC-K1)

9.4 The interaction term for a potential

Here the two terms are the kinetic one $T_1=-iH_0$ and the potential one $T_2=-iV$, with flows $U_0(t)$ and e^{-itV}; $E^{\mathrm{mult}}(t)=U(t)-U_0(t)e^{-itV}$ with $U=e^{-it(H_0+V)}$. The conjugation defect of the earlier frame is a different object.

Theorem 9.3 Estimate with optimal constant (ENC-A9)

For $V\in W^{2,\infty}$ real and $f\in H^1$, $t\geq 0$:

$$\|E^{\mathrm{mult}}(t)f\|\leq \tfrac{t^2}{2}\big(\|V''\|_\infty\|f\|+2\|V'\|_\infty\|f'\|\big)+\tfrac{2t^3}{3}\|V'\|_\infty^2\|f\|,$$

and the constant $2/3$ in the cubic term is optimal. For the companion variant (b2) the closed form is proved; the estimate of the first draft was false and the corrected one is proved. [proved, F8] (ENC-A9(a),(b1),(b2),(b4))

- NS-EQUIV: equivalence of the two frames on bounded Ω [proved, F8] (NS-EQUIV-Ω). FAD-1 [proved, F8]. FW-1: for free evolution [proved, F8]; with V, [from cited source].
- One earlier estimate was [false] as first written and replaced by a corrected one (audit).

Takeaway. A potential can be undone by an explicit Volterra operator at a known q (ENC-1), but there is no universal, V-blind encoder, and for a truly time-dependent potential every encoder must itself depend on time. The encoder is a change of frame, not a way to find V.

CHAPTER 10

A universal dictionary and misfit certificates

In this chapter

Starting from	Data $u \in H^m(\Omega)$, $\Omega \subset \mathbb{R}^{1+d}$ bounded Lipschitz (or $u \in C^\infty$). A *filtered dictionary* $\mathcal{D}_0 \subset \mathcal{D}_1 \subset \cdots$ of finite sets of operators ("atoms", with height function and finitely many atoms per height); $\mathcal{D}_\infty = \bigcup \mathcal{D}_n$. At level n: homogeneous fit means some $c \neq 0$ has $\sum_i c_i D_i u = 0$; misfit means none exists.
Why this chapter	The user asked for a dictionary that "takes everything into account". A literal reading is impossible; the strongest true form is: (i) a countable filtered dictionary is *universal by approximation* for each natural class of operators, (ii) exact universal identification is impossible, (iii) when the true law is not in the dictionary, a *certificate of misfit* can be issued, and (iv) when it is, identification in the limit holds.
Role in the volume	Eighth link; the dictionary supplies the data model of the expanded algorithm (Chapter 12). Status: Phase 8, audited with fixes ([proved, F8]); candidates are marked.



10.1 Universality: one class at a time

Class	What is proved	Status
U1: local linear operators	Every linear T on $C^\infty(\Omega)$ with $\operatorname{supp} Tu \subset \operatorname{supp} u$ is a differential operator of locally finite order (Peetre); hence $\{a(t,x)\partial^\alpha\}$ with a in a countable dense basis is universal on compacts. Singular atoms and L^p density proved; convergence rates only cited.	[proved, F8] (Peetre [from cited source], KMS 19.1) (DICT-U1, U1b)
U2: local nonlinear	Smoothness and Peetre–Slovák: polynomials in $(t, x, J^k u)$ are universal; proof avoids KMS 18.16.	[proved, F8] (DICT-U2)
U3(c): symbols	Fourier multipliers $\lvert\xi\rvert^s$, rational, log: dense graded classes; x-dependent case cited.	[proved, F8]/[from cited source] (DICT-U3c)
U3(d): memory kernels	Completely monotone kernels represented diffusively (Bernstein); uniform in t for $t^{-\beta}$.	[proved, F8] (DICT-U3d)
Delay	Delay equation as an augmented transport system; distance to a transport-free class exactly 1 (also on $(0,T)$).	[proved, F8] (DICT-N3)
Fractional	Caffarelli–Silvestre extension with explicit constant.	[proved, F8] (DICT-CS1)
U4: systems	Matrix versions via Chapter 7.	[proved, F8] (DICT-U4)

10.2 The negative theorem N1: what "everything" cannot mean

Theorem 10.1 N1: limits of any countable dictionary

(i) (Baire) In every infinite-dimensional Fréchet class of operators (e.g. coefficients $a \in C^\infty(\Omega)$ in $\partial_t - a\partial_x$) the operators representable *exactly* by a countable dictionary form a meagre set.

(ii) No finite data certifies fit or identification (jets, finitely many functionals, finitely many approximate data).

(iii) For closed-form data $u \in \mathrm{EP}$ ($N \geq 2$ variables) and any law $L \in \mathrm{Ann}(u)$ with $\deg L \geq 1$, the level-n Gram kernel exceeds the consequences of L for all large n (non-monotonicity).

(iv) Finite-output procedures: (a) in the string model the set of operators ever emitted (even as Gold limits) is countable, so for all but countably many c the operator $\partial_t - c\partial_x$ is never output; (b) with exact real outputs (BSS model) stabilisation in finite time is possible (Chapter 11); (c) whether $\mathrm{Ann}(u) \neq 0$ cannot be decided even in the limit.

(v) The generic u has no law in the dictionary, even nonlinear, in the sense of meagre and of prevalence.

[proved, F8] (DICT-N1)

An original claim was false

The first formulation of the last item (N1(vi)) was [false] (counterexample $u = t + x$ in the class $u_t = a(t,x)u_x$) and was replaced by the *rank theorem*: with a single coefficient and $\{\partial^{\alpha_1} u \neq 0\}$ dense the law is identified for every bound R'; with two dependent residual atoms (∂_x and 1) it can fail. [proved, F8] (DICT-RANGO)

10.3 Misfit certificates

When u is *not* in the span of level n, a finite object can prove it.

Theorem 10.2 Certificates MF1–MF3 and the fit/misfit asymmetry

- **MF1** (Hahn–Banach): $\operatorname{dist}(D_0 u, W) = \max\{|\varphi(D_0 u)| : \|\varphi\| \leq 1, \varphi|_W = 0\}$ with $W = \operatorname{span}\{D_i u\}_{i \geq 1}$; any admissible φ gives a lower bound.
- **MF2** (Gram, Hilbert case): $r_n^2 = \det G(D_0 u, \ldots, D_N u) / \det G(D_1 u, \ldots, D_N u)$; the optimal certificate is the normalised residual.
- **MF3** (weak form): independence of distributions is detected by finitely many test functions, $\det(\langle f_i, \psi_j \rangle) \neq 0$; only integrals of u are used; robust under noise by Weyl's inequality for singular values.
- **Theorem A**: the misfit set $U_n = \{u : D_0 u, \ldots, D_N u \text{ independent}\}$ is open and each of its points has a finite certificate; the fit set F_n is closed with empty interior, and even finitely many *exact* data (arbitrary continuous functionals) cannot certify fit for local atoms.

[proved, F8] (DICT-MF, DICT-A)

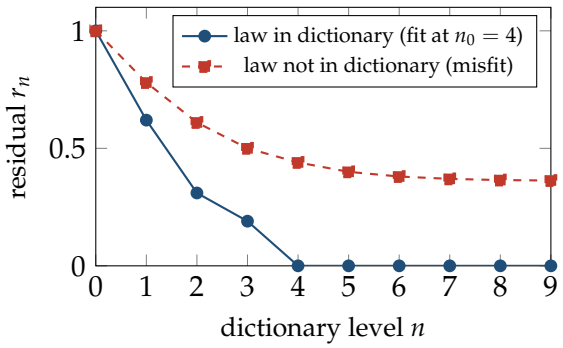


Figure 10.1: The residual $r_n = \min_{L \in \mathcal{A}_n} \|Lu\|$ is non-increasing. If the law is in the dictionary it reaches 0 at the minimal level n_0; otherwise it stays positive and the positive floor is itself the certificate of misfit (schematic values).

10.4 Best approximation and the rules R0–R6

- **B1–B2.** The least-squares coefficients c_n solve $G'c = -b$; if the Gram matrix of $D_1 u, \ldots, D_N u$ has smallest eigenvalue $\lambda_{\min} > 0$ and a candidate $\tilde{L}_n$ has residual $\eta = \|\tilde{L}_n u\|$, then $|c_n - \tilde{c}_n| \leq 2\eta / \sqrt{\lambda_{\min}}$. [proved, F8] (DICT-B2). Optimal-order statement via Kolmogorov widths: only cited, to be checked (B3).
- **Rules.** R0 filtration by height; R1 test at level n; R2 coherence (kernel must stay equal to the consequences of the accepted law); R3 exact gain of one atom, $r_{n+1}^2 =$

$r_n^2 - |\langle \mathrm{res}_n, Du\rangle|^2 / \|P_W^{\perp} Du\|^2$ (greedy orthogonal least squares, prior art); R4 change of family by residual signatures (heuristic only, no theorem); R5 stopping rule under noise; R6 change of frame by the Volterra encoder of Chapter 9. Theorem G-R: under exact data and uniqueness, R1 stops exactly at n_0. [proved, F8] (DICT-GR, DICT-R6)

- **I-LIM (identification in the limit).** If a unique law L with bounded coefficients exists and the dictionary approximates it, minimisers L_n of $\|L'u\|_{L^2}$ over the level (with bounded coefficients) converge to L with a modulus ω; with noise δ, $\|L^{\delta}_{n(\delta)} - L\| \leq \omega(C_u \varepsilon_n + 2C_{R'}\delta)$. Under the cone condition and a single coefficient with $|\partial^{\alpha_1} u| \geq \gamma > 0$ the rate is $O((\varepsilon_n + \delta)^{2/(N+2)})$. [proved, F8] (DICT-ILIM); for derivatives of order $s \geq 1$ it is a candidate (Gagliardo–Nirenberg to verify).
- **Exponential families.** Exact polynomialisation, transport of transcendence degree without Ax, kink and 2-kink identification in the exponential dictionary: [proved, F8] (NS-EXPALG); the 2-kink case checked on examples; Painlevé as a tool only.

Takeaway. *Fit* is a closed, nowhere dense condition: no finite amount of data can ever establish it. *Misfit* is an open condition: a finite certificate exists. The dictionary strategy therefore reports one of two honest outcomes: a law, identified in the limit with a rate, or a certificate that no law at the current level fits, together with the best approximation and its error bound.

CHAPTER 11

Convergence, noise, locality, dimension, boundary

In this chapter

Starting from	Data: the Taylor jet of an analytic u at a point (exact data, jets $j_K u$ for $K = 0, 1, 2, \dots$), or u with noise $\|\tilde{u} - u\| \le \delta$ in H^d or L^2; $\Omega \subset \mathbb{R}^n$, $n = 1 + d'$ open connected. Level m of the dictionary: $\mathcal{D}_m = \{\partial^\alpha : |\alpha| \le m\}$, $\Phi_m(u)\lambda = \sum \lambda_\alpha \partial^\alpha u$.
Why this chapter	Chapter 10 said what a dictionary can promise. Here we say what happens as the amount of data (or of dictionary) grows, when data are noisy, when the law may be non-local, in higher dimension, with boundary conditions, and for the splitting schemes used to compute E_{int}.
Role in the volume	Last theoretical chapter; its contract (CONV-T7) fixes what each step of the expanded algorithm in Chapter 12 may promise. Status: Phase 8, audited with fixes.



11.1 Identification in the limit

Define, for the jet of order K, the kernel $N_{m,K} = \{Q \in \mathcal{R}_{\le m} : \partial^\beta(Qu)(p_0) = 0,\ |\beta| \le K - m\}$, and the *minimal-degree rule*: $m^*(K) = \min\{m \le K : N_{m,K} \ne 0\}$, output $O_K = \gcd N_{m^*(K),K}$.

Theorem 11.1 Identification in the limit (CONV-T1)

Under analyticity, with $d = \min\{\deg Q : 0 \ne Q \in \mathrm{Ann}(u)\}$:

(1) for $K \ge K^*$: $m^*(K) = d$ and $O_K = \gcd(\mathrm{Ann}(u) \cap \mathcal{R}_{\le d})$;

(2) if $\mathrm{Ann}(u) = (P)$ then $O_K = P$ for all $K \ge K^*$, without knowing $\deg P$;

(3) if $\mathrm{Ann}(u) = 0$ then $m^*(K) \to \infty$;

(4) with a bound B and $\mathrm{Ann}(u)$ generated in degree $\le B$: $O_K^B = \mathrm{Ann}(u)$ eventually;

(5) the sequence (O_K) takes at most $(d+1)(d+2)/2$ values, for every u;

(6) K^* is not bounded by any function of d: $u = 1 + x^{N+2}e^{x^2}$ has $\mathrm{Ann} = (\tau)$, $d = 1$ and $K^* = N + 2$ (output $O_K = 1 \ne \tau$ for $1 \le K \le N + 1$).

[proved, F8] (CONV-T1)

Theorem 11.2 Analytic Gold theorem: no learner can decide "a law exists" (CONV-T2)

For any sequence of maps $\varphi_K : \mathbb{R}^{J_K} \to \{\text{law}, \text{no law}\}$ (no assumption of computability, continuity or measurability) there is an entire function u whose jets make $\varphi_K(j_K u)$ fail to converge to the correct answer. The proof uses $g_0 = e^{|z|^2}$ (for which $\mathrm{Ann}(g_0) = 0$, avoidance lemma) and polynomials, which always satisfy some law. [proved, F8] (CONV-T2)

Takeaway. The correct law is eventually reproduced by a finite computation (the stabilisation level K^*), but nobody can know when. That is exactly the content of Gold's learning-in-the-limit: verifiable in the limit, not decidable in the limit.

11.2 Noisy data

Statement	Meaning	Status
$\{u : \mathrm{Ann}(u) = 0\}$ and $\{u : \mathrm{Ann}(u) \neq 0\}$ are both dense in any reasonable function space (polynomials dense, $e^{\lvert z\rvert^2} \in X$)	Identification is discontinuous everywhere (CONV-T3).	[proved, F8]
Conditional stability at a level: if $\sigma_2(\Phi(u)) \geq \rho > 0$ (margin) and $\lVert\tilde{u} - u\rVert_{H^{n_0}} \leq \delta$ then $\sin\angle(\tilde{\lambda}, \lambda^*) \leq 2\delta/\rho$; order δ/ρ is sharp (CONV-T4)	Stable *relative to a level, with a margin*.	[proved, F8]
Consistency with a discrepancy rule: $m(\delta) = \min\{m : \sigma_1(\Phi_m(\tilde{u})) \leq \delta_m\}$; correct output below a threshold $\delta_0(u)$ not computable from the data (CONV-T5)	The price of noise: the threshold is invisible.	[proved, F8]

11.3 Locality

When is a law a *differential* operator rather than non-local? For $u = \int e^{a(\kappa)t + i\kappa x}\hat{f}(\kappa)\,d\kappa$ with $\hat{f}$ compactly supported:

Theorem 11.3 Spectral criterion at the level of the datum (CONV-T6)

$\mathrm{Ann}(u) = I(\Gamma_E) = \{Q : Q(a(\kappa), i\kappa) = 0 \text{ for a.e. } \kappa \in E\}$, $E = \{\hat{f} \neq 0\}$. On an interval where a is real-analytic, $\mathrm{Ann}(u) \subseteq I_J$ with I_J prime: it is 0 if $a|_J$ is transcendental over $\mathbb{C}(\kappa)$ and principal otherwise. If $a|_E$ is algebraic, the local law and the non-local law are indistinguishable from u. [proved, F8]

Irrational fractional powers $a = -\kappa^\alpha$ and the Lambert-W delay symbol give $\mathrm{Ann}(u) = 0$ [proved, F8] (CONV-P6): genuinely non-local. But the delay $u_t(t,x) = u_{xx}(t-\sigma, x)$ becomes a *local* transport system in (t, θ, x) with boundary coupling (class L^+). A three-way classification: local L; local after augmentation $L^+_{\mathcal{K}}$; genuinely non-local $N_{\mathcal{K}}$; and without a declared class the question is vacuous [proved, F8] (CONV-P7).

11.4 Dimension $1 + d'$

Reduction to the scalar case is dimension-independent [proved, F8] (CONV-P8). In $n \geq 3$ variables two non-comaximal factors have an *infinite-dimensional* common kernel (their variety has positive dimension), so the factor-splitting ambiguity is infinite in dimension when $n \geq 3$ and nonzero: $\mathcal{K} = 0$ iff the factors are pairwise comaximal [proved, F8] (CONV-D1). Existence of the splitting on convex Ω in C^∞ [from cited source] (Malgrange; the $\mathcal{D}'$ case excluded and to be checked) (CONV-D2).

11.5 Boundary conditions

- **Invisible.** $\mathrm{Ann}(u)$ for analytic u is determined by the germ at any interior point: it does not depend on boundary data. Example: on $(0, L)$ the Dirichlet mode $e^{-\kappa_0 t}\sin \mu x$ and the Neumann mode $e^{-\kappa_0 t}\cos \mu x$ have the same annihilator $(\tau + \kappa_0, \xi^2 + \mu^2)$ [proved, F8] (CONV-B1).
- **Stability for a dictionary of traces** [proved, F8] (CONV-B2).
- **Tangential obstruction.** Above the minimal level the boundary margin is zero: for a flat boundary piece Γ transversal to t, if $Q^*(\partial)u$ has zero trace on Γ then so has $\tau Q^*(\partial)u$, hence $\sigma_2(\Phi_\Gamma(u)) = 0$ [proved, F8] (CONV-B3).
- **Periodic boundary as a boundary case.** On $I \times \mathbb{T}$, $\ker(fg) = \ker f + \ker g$ iff there is no *resonant* k (with $f(\cdot, ik)$, $g(\cdot, ik) \not\equiv 0$ and common roots); the earlier criterion without the clause on lines is false (example $f = \xi$, $g = \tau$) [proved, F8] (CONV-R1).

11.6 Splitting schemes with a boundary

For the advection–diffusion pair on $(0, L)$ with Dirichlet conditions, linearised splitting errors for modal data are computed for Lie and Strang splitting; a first-order-in-ε answer to the open question CONV-Q6 [proved, F8] (L^2), with first order in L^∞ for Strang checked on examples [checked on examples only] (CONV-T8).

11.7 Mori–Zwanzig

At constant coefficients, eliminating components creates no genuine non-locality: a square non-degenerate system gives every component the local law $\det D$ (MZ1); the exact Mori–Zwanzig form (Markov + memory + noise) is equivalent to $\det(\tau - A(\xi))(\partial)u = 0$ (MZ2). The criterion "$E_{\mathrm{int}} = 0 \iff [\mathcal{P}, L] = 0$" read as "memory zero $\iff [\mathcal{P}, A] = 0$" is [false] (take $a_{21} = 0, a_{12} = 1$). [proved, F8] (CONV-MZ1–MZ3)

11.8 The convergence contract for the algorithm

Each step of the algorithm is labelled by what it can promise.

Label	Meaning	Examples
(A)	Finite and exact	Elimination on $\mathcal{D}_n$; global Gauss/Wei–Norman iff Wronskians $\neq 0$; products of flows for solvable Lie algebras; E_{int} in class $\mathfrak{B}$ (2^N terms); commuting-multiplier splitting; MZ1 elimination; Bézout construction on $\mathbb{T}$ for non-resonant modes.
(B)	Convergent with explicit bound	Dyson–Phillips series with bounded perturbation (error $Me^{\omega t}\sum_{k>N}(M\|B\|t)^k/k!$); Magnus for real matrices with $\int \|A\| < \pi$ (Moan–Niesen) and for bounded Hilbert operators (Casas); Lie/Strang under Trotter–Kato–Chernoff.
(C)	No guarantee	Magnus/BCH with unbounded generators and no hypotheses; E^{mult} for non-solvable algebras; splitting with a boundary beyond first order.
(Lim)	Correct from some step on, not certified	Identification (CONV-T1), choice of level (CONV-T5).

Proposition 11.4 Composition (CONV-T7)

If step i is computed with error $\leq \varepsilon_i$ on the computed input and later steps are L_j-Lipschitz, the total error is $\leq \sum_i \left(\prod_{j>i} L_j\right)\varepsilon_i$. [proof written, not audited]

A guaranteed chain contains only steps (A) and (B); a step (C) is outside the guarantee and a step (Lim) is exact only "under the event $K \geq K^*$", which must be declared to the user.

CHAPTER 12

The expanded algorithm

In this chapter

Starting from	All of Chapters 3–11. The data model is one of: DM1, a finite exact list (as M1 of Chapter 3); DM2, exact jets of an analytic u at a point (computation in the BSS real model); DM3, a function $u \in H^m(\Omega)$, possibly with noise; DM-φ, u known up to a phase $e^{i\theta(t)}$.
Why this chapter	The user's original request was an algorithm, with the research as its proof of correctness. This chapter assembles the steps and says which theorem guarantees which step, under which hypotheses, and what is *not* guaranteed.
Role in the volume	Synthesis of the volume. Chapter 13 gives the status of every ingredient.



What this chapter is and is not

The research plan has a Phase 9 whose output would be a document "ALGORITMO_ESTESO" with the specification reviewed adversarially. That document does *not* exist yet (checked on disk). The steps below are my assembly from the proof packages (the dictionary rules R0–R6, the convergence contract CONV-T7, and the theorems of Chapters 3–11). It is a faithful map of the proven pieces, but it has not been red-teamed as a specification. Treat it as a reading guide, not as a certified algorithm.

12.1 The flow

12.2 Step by step

Step	Does	Guarantee, hypotheses	Source	Label
0	Fix the data model and the domain Ω.	Topology of Ω matters: simply connected for factor splitting; bounded Lipschitz for H^m.	Ch. 4	—
1	Enumerate the filtered dictionary by height.	Universality class by class; no exact universal dictionary exists (N1).	DICT-U1–U4, N1	[proved, F8]
2	Test fit at level n via the Gram matrix (exact data) or the weak form (noisy).	Fit certified only in the limit; misfit has a finite certificate; soft threshold under noise.	MF1–3, Thm A, CONV-T5	(Lim)

Step	Does	Guarantee, hypotheses	Source	Label
3	Identify: compute $\mathrm{Ann}(u) \cap \mathcal{R}_{\le n}$, then G_u, J_u by the EP test; for systems use the socle criterion; modulo gauge.	Decidable with a bound B; black-box impossible; sums of exponentials give finite certificates.	C3-T1, T6; P4-T4, T5; SYS-EX	(A) with B; (Lim) otherwise
4	If a potential V is present and known: apply the Volterra operator W (frame change).	Exists at known q on a symmetric interval; no blind or time-independent encoder for truly time-dependent V.	ENC-1, ENC-P0, ENC-N1	(A)
5	Split per factor (simply connected Ω); within each factor compute E_{int} (closed form, class $\mathfrak{B}$); nonlinear branch: jet expansion and dictionary relative identification.	Exact; cost $O(N)$ per value; *not* stable in H^m; closed form only before shocks, $\Theta(\nu)$ in L^1 after.	C12-T1′, P1-T2, P1-T3; S4T, S4ID	(A)/(B)
6	Attach to every output the label of the weakest step used and the event under which it holds.	Composition bound.	CONV-T7	—

12.3 What the output can honestly claim

1. A law L with a guarantee "L is the unique law of height $\le n_0$" only *under the event* $K \ge K^*$, which is not certified; or
2. a certificate of misfit at the current level, together with the best approximation and its error bound (B2: $|c_n - \tilde{c}_n| \le 2\eta/\sqrt{\lambda_{\min}}$);
3. components per factor, with the finite exponential-polynomial ambiguity written out;
4. E_{int} as a closed form where the commuting/class-$\mathfrak{B}$ hypotheses hold, otherwise a series with a Dyson/Magnus remainder, otherwise no guarantee;
5. for systems and potentials: results modulo gauge and modulo constants (the phase e^{-ict}).

Takeaway. The expanded algorithm is a decision procedure with three honest exits: "law found (eventually)", "no law at this level (certificate)", and "no guarantee for this step". What it cannot do is declare success from finite data, nor a stable per-term decomposition in Sobolev norms.

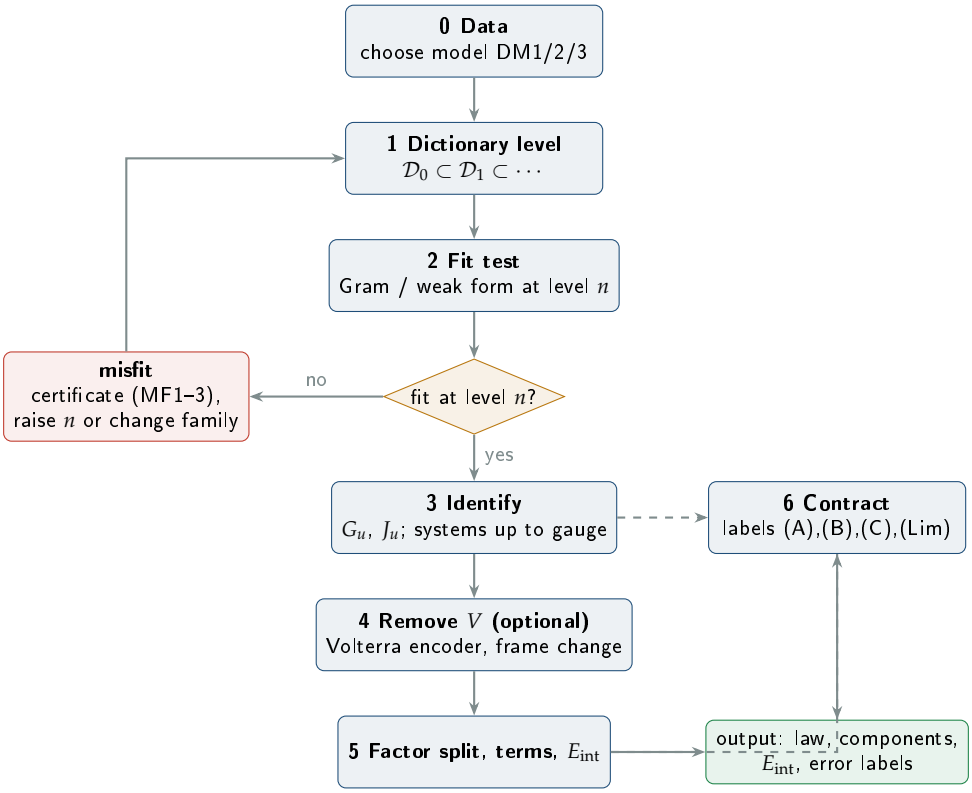


Figure 12.1: Flow of the expanded algorithm. The red branch (misfit) is a legitimate output: a certificate that the dictionary level cannot represent the law.

CHAPTER 13

Status, gaps, and what remains

In this chapter

Starting from	Everything in Chapters 2–12.
Why this chapter	A theory is only as useful as its stated limits. This chapter collects, in one place, the scoreboard, the independent checks, the formalised core, the open problems, the false statements the programme caught, and the part of the plan that has not been done.
Role in the volume	Closes the volume. It is the chapter to read before relying on any result.

13.1 Scoreboard

Phases	Source of the proofs	State
1–3 (linear theory)	8 proof packages, 11 audit rounds	Core theorems proved and checked; negative results proved; some questions closed later.
4 (nonlinear)	3 proof packages, audits S-CHECK-9..11	Core proved; shock and KdV asymptotics proved modulo cited facts.
5 (integration, Lean)	integration session, Lean 4	No theorem fell; 15 algebraic statements formalised.
5b (open-problem sweep)	OP-ALG, OP-ANA, OP-FUN, OP-NL, OP-R2, OP-F5A–D	Most questions closed, some partially (Section 13.6).
6–7 (scouting, setup)	scouting reports, synthesis	Closed; gate auto-approved by delegation, revocable.
8 (systems, variable coefficients, encoder, dictionary, convergence)	5 proof packages, audit S-CHECK-F8	Proved with fixes ("proposed state"); ledger registration pending.
9 (expanded algorithm)	—	**Not done**: no specification document exists; Chapter 12 is my assembly.
10 (integration of Phase 8, Lean for Phase 8)	—	**Not done**.
11 (final paper and audit)	—	**Not done**; this pair of volumes is not that deliverable.

13.2 What "independently checked" meant

For every proof package a separate reviewing session (a fresh AI agent without access to the author's reasoning) re-derived or attacked the results, producing a list of fixes tagged by severity; the fixes were then applied (the packages carry the fix markers). For the principal theorems an additional *kill-argument* was run: one agent tries to refute, another adjudicates. In the linear phases some kill-arguments were run by two fresh agents instead of an external model. Errors found and corrected by the reviewers include: the constant of a Pearcey integral ($P = 1.188654$), a sign error at $n = 0$ in a Lax-pair statement, a false lemma ($g \in C^1$ instead of C^2), a gap in a unique-continuation step, a false statement "crossing implies pole" for derogatory matrices, and the whole first draft of the blind-encoder claim.

Limits of this assurance

These are AI reviewers, not human referees. Some citations were verified only on secondary text or in abstract (the research files list them). Computer-algebra checks verify examples, not theorems.

13.3 The formalised core (Lean 4 / Mathlib)

A Lean project builds (9056 jobs) with no `sorry` and no added axiom; all 15 theorems depend only on `propext`, `Classical.choice`, `Quot.sound`, and an independent reviewer compared each formal statement with its informal version without disagreement. Formalised: the algebraic kernel of the exact formula for the additive interaction term with commuting flows (R5-T2(i)); the comaximal decomposition with explicit projector over any commutative ring and module (II-T3) with its equivalence "polynomial projectors iff disjoint varieties" via the Nullstellensatz; the L_0 level of R2-T2; reducibility of the quadratic factor and irreducibility of $\tau - a$ (C12-P2, P3, promoted to proved); normality of the companion matrix for $k = 2$ (II-P5). **Not** formalised: C3-L3(b) (needs a function-space library), and anything analytic (distributions, semigroups, Sobolev spaces, Ehrenpreis); Mathlib lacks the libraries.

13.4 Textbook sanity checks

All results were also confronted with standard examples (heat, advection–diffusion, heat$\times$transport, Airy, variable coefficients, radial Bessel): 150 of 150 symbolic checks agreed, and no conflict between results was found (EX-CAL, EX-AD, EX-CT, EX-AIRY, EX-VAR, EX-BES). Bessel radial: $u_G = t^{-1}e^{-x^2/4t}$ identifies its operator (EX-BES, proved).

13.5 Statements the programme proved false

Statement (as first believed)	Where refuted
Per-term splitting $\ker(L_1+L_2)\cap(\ker L_1+\ker L_2)=0$	R2-T3
Continuity of E_{int} in H^m	P1-P1(e)
Regularisation can remove $E_{\mathrm{int}}^{\perp}$	P2-T2, P2-T4
Decomposition preserves H^m	C12-C1
Factor decomposition on domains with holes	C12-P10
Finite certificates of identification	C3-P1; P4-T5
Restriction to a variable-coefficient module identifies	P4-C1
Finite product of term flows equals the full flow (infinite Lie algebra)	P1-T5, S4T-C1
"Spectrum of the expansion on a bounded domain is intrinsic"	C3-C1
$\sigma_p=\emptyset$ implies identification	P4-T2 (counterexample e^{x^2})
Blind encoder for a class of potentials	ENC-N1
"Exceptions change the time" in time-dependent encoders	ENC-P0 (first draft)
Every $u\not\equiv 0$ identifies $V(t,x)$	VAR-T1(c)
Original N1(vi) (rank statement)	DICT (counterexample $u=t+x$)
Memory zero iff $[\mathcal{P},A]=0$	CONV-MZ3
"Crossing implies pole" for derogatory A	II-T2(b′)
Reflection-free NLS without multiplicity	OPF5D
Infinite analyticity radius for Burgers (cited preprint)	KIEFFER-C

13.6 Open and conditional

ID	Question	State
INT-Q1	KdV with analytic data, nonzero reflection: autonomous ODE in x?	[open]; three proved obstructions
OPA-Q1 general	Uniform shock rate with several shock births	[conditional] (localisation lemma)
OPA-Q2 density	Density of the Morse+(SB) class	sketch
OPR2-Q2′	Wedges with degenerate directions	partial (tangent model closed)
OPR2-Q3	Repeated socle and "no finite test"	partial (under two assumptions)
OPF5C-Q1	$\delta_N(K)$ for general N, $N<K<2N-2$	[open] (three conjectures)
S4ID-P8, OPR2-P7	Asymptotics with derivatives	[conditional] (uniformity in x)
INT-T1(B)	Hirota N-solitons without reflection	[conditional] (primary source not located)
Density off convex sets	Closed sum of kernels dense	[open]

ID	Question	State
P4-Q-min	Minimal bound for decidability	[open]
C3-Q2 general	Intrinsic criterion in A_2 for order $m \geq 2$, non-monic	[open]
II-T2(b′)	Corrected statement for derogatory A	[open]
Q-S8SYS-1	Finite flow products with polynomial times	[open]
VAR-P12 general, VAR-9 converse	Floquet differential logarithm; continuous choice of a killing matrix	[open]
DICT-U3d rate, ILIM-Hs, B3	Rates and optimality statements	candidate / cited
CONV-Q17	Splitting in $\mathcal{D}'(\Omega)$ on convex Ω	to verify

13.7 What these volumes do not contain

- **Everything of Phase 9–11**: an audited specification of the expanded algorithm, the integration and formalisation of Phase 8, the final paper.
- **Full proofs** of the following (their proofs are in the research files, listed in Volume II, Chapter 12): shock asymptotics, KdV soliton degree and syzygy tables, scattering identification, the convex-domain density theorem, unique continuation and Carleman estimates, differential Galois statements, the Weyl-algebra criterion, CONV-T1 and T2 proofs, most of the dictionary universality proofs, the boundary and splitting analyses.
- **The equation catalogue** of about 60 types (research reports PR-C2a–e): used as scaffolding, not reproduced.
- **Phase 8 ledger registration**: results appear here with the states proposed by the auditors, not yet written in the ledger.

Takeaway. Treat [proved] as "checked by a fresh AI reviewer with fixes applied", [proof written, not audited] as "plausible, not checked", [from cited source] as "published, source checked", and everything under [open], [conditional], [checked on examples only] as *not established*. The programme's most reliable output is the *pattern of impossibility*: what no finite data, no regularisation, no blind encoder and no countable dictionary can do.

Sources cited

Only sources that are named in the text are listed, with the details recorded in the project's own verified bibliography (`BIBLIOGRAFIA.md`, 112 sources, of which 34 have the theorem number checked and 75 the content checked; the others are metadata only). Where the research files record that a number or a page is not verified, it is omitted here.

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